Working on Real Project with Python

(A part of this Data Analysis)

The Flights Dataset

```
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import pandas as pd
import seaborn as sns
pd.set_option('display.max_columns', 50)
df = pd.read_csv('/kaggle/input/flight-delay-prediction/Flight_Delay.csv')
df.shape
df.head(10)
```

```
(1000, 10)
```

	Airline	Flight	Airport_Dep_Code	Airport_Dep_City	Airport_Arr_Code	Airport_Arr_City	Date	Scheduled_Dep_Time	Scheduled_Arr_Time	Delay
0	Spanair	5413	CDG	Cheng_Jeu	BCN	Barcelo	Monday	2.17	3.55	1.000
1	Spanair	5413	CDG	Cheng_Jeu	BCN	Barcelo	Monday	2.15	3.55	1.000
2	Iberia	3179	CDG	Cheng_Jeu	Cuba_Havana	Cuba_Havana	Monday	2.15	5.45	1.000
3	Iberia	3179	CDG	Cheng_Jeu	Cuba_Havana	Cuba_Havana	Monday	2.15	5.45	1.000
4	Iberia	3179	CDG	Cheng_Jeu	Cuba_Havana	Cuba_Havana	Monday	2.15	5.45	1.000
5	Iberia	3179	CDG	Cheng_Jeu	Cuba_Havana	Cuba_Havana	Monday	2.15	5.45	1.000
6	Iberia	3179	CDG	Cheng_Jeu	Cuba_Havana	Cuba_Havana	Monday	2.15	5.45	1.000
7	Iberia	3179	CDG	Cheng_Jeu	Cuba_Havana	Cuba_Havana	Monday	2.15	5.45	1.000
8	Iberia	3179	CDG	Cheng_Jeu	Cuba_Havana	Cuba_Havana	Monday	2.15	5.45	1.000
9	Iberia	3179	CDG	Cheng_Jeu	Cuba_Havana	Cuba_Havana	Monday	2.15	5.45	1.000

```
(8219, 10)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8219 entries, 0 to 8218
Data columns (total 10):
 #   Airline             8219 non-null   object
 #   Flight              8219 non-null   object
 #   Airport_Dep_Code    8219 non-null   object
 #   Airport_Dep_City    8219 non-null   object
 #   Airport_Arr_Code    8219 non-null   object
 #   Airport_Arr_City    8219 non-null   object
 #   Date                8219 non-null   object
 #   Scheduled_Dep_Time  8219 non-null   float64
 #   Scheduled_Arr_Time  8219 non-null   float64
 #   Delay               8219 non-null   int64
dtype: float64, int64, object
memory usage: 642.2+ KB
```

```
df.describe()
```

	Scheduled_Dep_Time	Scheduled_Arr_Time	Delay
count	8.219.000	8.219.000	8.219.000.000
mean	12.214.167	15.347.711	2.008.148.000
std	3.463.000	3.463.000	1.000.000.000
min	0.000.000	0.000.000	0.000.000.000
25%	11.000.000	13.750.000	1.000.000.000
50%	12.250.000	15.500.000	2.000.000.000
75%	13.750.000	16.830.000	3.000.000.000
max	19.750.000	22.830.000	4.000.000.000

```
df.isnull().sum()
```

```
Airline             0
Flight              0
Airport_Dep_Code    0
Airport_Dep_City    0
Airport_Arr_Code    0
Airport_Arr_City    0
Date                0
Scheduled_Dep_Time  0
Scheduled_Arr_Time  0
Delay               0
dtype: int64
```

```
df.tail(10)
```

	Airline	Flight	Airport_Dep_Code	Airport_Dep_City	Airport_Arr_Code	Airport_Arr_City	Date	Scheduled_Dep_Time	Scheduled_Arr_Time	Delay
8209	Air_Europa	1.673	Cancun	Cancun	Mad_Barajas	Madrid	Friday	19.00	6.05	1.000
8210	Air_Europa	1.673	Cancun	Cancun	Mad_Barajas	Madrid	Friday	19.00	6.05	1.000
8211	Air_Europa	1.673	Cancun	Cancun	Mad_Barajas	Madrid	Friday	19.00	6.05	1.000
8212	Air_Europa	1.673	Cancun	Cancun	Mad_Barajas	Madrid	Friday	19.00	6.05	1.000
8213	Air_Europa	1.673	Cancun	Cancun	Mad_Barajas	Madrid	Friday	19.00	6.05	1.000
8214	Air_Europa	1.673	Cancun	Cancun	Mad_Barajas	Madrid	Friday	19.00	6.05	1.000
8215	Air_Europa	1.673	Cancun	Cancun	Mad_Barajas	Madrid	Friday	19.00	6.05	1.000
8216	Air_Europa	1.673	Cancun	Cancun	Mad_Barajas	Madrid	Friday	19.00	6.05	1.000
8217	Air_Europa	1.673	Cancun	Cancun	Mad_Barajas	Madrid	Friday	19.00	6.05	1.000
8218	Air_Europa	1.673	Cancun	Cancun	Mad_Barajas	Madrid	Friday	19.00	6.05	1.000

```
df.shape
```

```
(8219, 10)
```

```
df.dtypes
```

```
Airline             object
Flight              object
Airport_Dep_Code    object
Airport_Dep_City    object
Airport_Arr_Code    object
Airport_Arr_City    object
Date                object
Scheduled_Dep_Time  float64
Scheduled_Arr_Time  float64
Delay               int64
dtype: object
```

```
df.nunique()
```

```
Airline             10
Flight              10
Airport_Dep_Code    10
Airport_Dep_City    10
Airport_Arr_Code    10
Airport_Arr_City    10
Date                 7
Scheduled_Dep_Time  10
Scheduled_Arr_Time  10
Delay               10
dtype: int64
```

```
df.sample(10)
```

	Airline	Flight	Airport_Dep_Code	Airport_Dep_City	Airport_Arr_Code	Airport_Arr_City	Date	Scheduled_Dep_Time	Scheduled_Arr_Time	Delay
7.0000	Iberia	3.179	Cancun	Cancun	Mad_Barajas	Madrid	Friday	19.00	6.05	1.000